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Original Research

A Tactile Trans-Flexor Carpi Radialis Surface Landmark Technique for Median Nerve Local Injection: A Cadaveric Study

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Purpose: Median nerve (MN) block under local anesthesia at the wrist is commonly performed but can be limited by anatomic variation, particularly absence of the palmaris longus tendon. This study investigates a novel trans-flexor carpi radialis (trans-FCR) technique for MN anesthesia that uses the flexor carpi radialis (FCR) tendon as a consistent surface landmark.
Methods: Fourteen fresh frozen upper extremity cadaver specimens were used. A 25-gauge needle was inserted through the FCR sheath approximately 4 cm proximal to the wrist crease in a transverse fashion, angled approximately 0° to the vertical and parallel to the wrist crease. The trans-FCR technique was tested against two other standard surface landmark techniques.
Results: The trans-FCR technique avoided penetration of critical structures such as the MN (3.5 mm), radial artery (8.5 mm), and palmar cutaneous branch of the MN (3.8 mm). The MN crossed under the long finger flexor digitorum superficialis at an average of 4.0 cm proximally from the wrist crease with a range of 2.8–4.8 cm. Three specimens had absent palmaris longus. There was no difference in injury rate to surrounding critical structures regardless of palmaris longus presence. No incidences of MN penetration occurred with the trans-FCR approach.
Conclusions: The trans-FCR technique is an anatomically based method for MN block under local anesthesia that uses a consistent tactile landmark and without dependence on the palmaris longus tendon. Clinical studies are needed to evaluate its efficacy in vivo.
Type of study/level of evidence: Basic Science Anatomic Research V.

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Local anesthesia of the median nerve (MN) has been a procedure often used by surgeons, hospitalists, and emergency room physicians. Peripheral nerve block of the MN inhibits impulse transmission through blockage of cell membrane-based voltage-gated sodium channel influx, thereby preventing depolarization.[1] However, the accurate depositing of local anesthetic in the proximity of the nerve is essential to the successful termination of pain signals in the cortex.[2,3] This impulse blockade can last hours to days depending on medication used and block technique.[4] The predictable and reproducible nature makes local anesthesia a staple for bedside and surgical hand procedures.[5,6] Despite its efficacy, local anesthetics are not without their risks, including direct needle trauma, intraneural injection, hematoma from puncturing nearby vessels.[7] In rare circumstances, life threatening complications such as seizure, altered mental status, dysrhythmias, hypotension, respiratory failure, and cardiac arrest may occur after arterial infiltration causing central side effects.[1,8–10] At high concentrations, injury to local structures and nerves through direct cytotoxic damage, ischemic changes via vasoconstriction, and axonal nerve degeneration are possible but rare.[11–15]

The MN is a mixed nerve with both sensory and motor components. As the MN enters the wrist, it branches off into the sensory palmar cutaneous nerve, whereas the remaining MN contains the sensory fibers for the hand. Because of variability in anatomical landmarks, such as the palmaris longus (PL) tendon, localization of the MN can be difficult.[16]

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Traditional techniques for administering a MN block often involve using anatomical landmarks such as the PL tendon or flexor carpi radialis (FCR) tendon to guide needle placement. These methods are widely practiced because of their simplicity and the relatively high success rate in achieving effective anesthesia. However, the reliance on these anatomical landmarks presents limitations, particularly in patients with atypical anatomy or in cases where the PL tendon is absent, which occurs in approximately 15% of the population.[16] In such situations, the accuracy of nerve localization can be compromised, leading to incomplete anesthesia or the need for multiple attempts, increasing patient discomfort and the risk of complications. These challenges underscore the need for alternative techniques that offer more consistent and reliable results across diverse patient anatomies.

Although traditional MN block techniques are generally considered safe and permanent neurologic injury is rare, transient needle contact with the MN can still cause sudden, severe pain even without clinically significant intraneural injection. Such pain may provoke reflexive hand withdrawal, creating discomfort for the patient and making the procedure less controlled for the operator. In contrast, passage through the FCR tendon and its sheath appears to produce milder, more predictable discomfort and may offer useful tactile feedback, as distention of the FCR sheath can guide depth correction. These observations and preliminary clinical experience prompted investigation of an alternative approach that may improve procedural consistency and patient comfort. However, this study is anatomical and does not include formal clinical outcome data.

This anatomical study describes a novel trans-flexor carpi radialis (trans-FCR) technique for MN block that uses the FCR tendon and sheath as a reliable tactile landmark, particularly when the palmaris longus (PL) tendon is absent.

Materials and Methods

This was a cadaveric study conducted using 14 fresh frozen upper extremity specimens to evaluate the feasibility and safety of the trans-FCR technique. Limbs were single limb donations that were independently collected. No limb had a contralateral extremity that was also included in this study. Dissection was carried down to the level of the needle tip to expose neurovascular structures under 2.5× loupe magnification. Measurements were taken from the needle tip to the nearest neurovascular structures using calipers. Measurements were taken to three decimal places; final measurements were reported to one decimal place. Aggregated measurements were reported as means ± SD. The distances from the needle to critical structures were collected, including the MN, radial artery (RA), and palmar cutaneous branch of the median nerve (PCBMN). Two investigators took measurements independently; instances of discrepancy were averaged. There was a maximum discrepancy of 0.1 mm across all measured distances. Measurements were analyzed to determine the safety and effectiveness of the trans-FCR technique compared to traditional techniques.

Trans-FCR technique

Application of the technique first involved placement of the forearm into a supinated position with the volar wrist pointing upwards and the wrist in neutral position (no ulnar deviation). The FCR tendon was then palpated along its length from the level of the wrist crease to the distal third of the forearm. The scaphoid tubercle may also be used as a distal landmark to locate the FCR tendon. A 25-gauge needle was inserted approximately 4 cm proximal to the wrist crease in a transverse orientation, with the needle angled approximately 0° to the vertical and parallel to the floor. When looking proximally to distally, the needle insertion was parallel to the wrist crease (Fig. 1). The needle was inserted until a tactile pop through the FCR tendon and the tendon sheath, confirming its position at the adjacent to the MN, proximal to the wrist crease (Fig. 2A).

Comparison techniques

The trans-FCR technique was used in comparison against two traditionally described techniques for MN injection that also avoid iatrogenic MN injury. We chose to use other techniques that also do not depend on the presence of palmaris longus.

The first technique (Fig. 2B) involved insertion of the 25-gauge needle in line with the fourth digit, at a location 1 cm proximal to the wrist crease technique. The needle was inserted in a 30° angle to the horizontal in a longitudinal orientation.

The second technique (Fig. 2C) involved insertion of the 25-gauge needle at a location 1 cm proximal to one-third the width of the wrist crease technique. The needle was inserted at a 30° angle to the horizontal and a longitudinal orientation to target the MN.

Statistical analysis

Data analysis was conducted using statistical software. Descriptive statistics included male/female, hand laterality, neurovascular injury, and presence of PL. Kruskal–Wallis test and other nonparametric statistical tests were applied to determine significant differences between the three tested approaches and their distance to critical structures such as MN, RA, and PCBMN. Chi-square tests were used to test categorical values. All statistical tests were two-tailed, and the significance level was set at 0.05.

Results

A total of 14 fresh frozen upper extremity specimens were dissected. The average age was 61, with a range of 29 to 81 years. There were six female specimens and eight male specimens. There were six left arms specimens and eight right arm specimens. There were 13 specimens that were identified as Caucasian or Hispanic, and one specimen was identified as African American. The average skin thickness was 2.7 ± 1.3 mm. The average wrist circumference was 17.5 ± 1.1 cm. Three specimens did not have PL.

The trans-FCR is a feasible approach that avoided penetration of critical structures, with the needle being the closest to the MN at an average distance of 3.5 ± 1.2 mm (Table 1). Presence or absence of the PL did not significantly change the distance of the needle insertion site to the RA ($P = .661$) or PCBMN ($P = .072$), while also maintaining good proximity to the targeted MN ($P = .171$). Needle insertion did not result in any instances of landing superficial to the antebrachial fascia.

The average MN diameter was 6.1 ± 0.7 mm, and the average ulnar nerve diameter was 3.3 ± 0.6 mm. The crossover location where the MN passed under the long finger flexor digitorum superficialis (L-FDS) was an average of 4.0 ± 0.6 cm (range 2.8–4.8 cm) from the wrist crease. In specimens with PL, the average crossover location was 4.0 ± 0.6 cm. In contrast, in specimens without PL, the average was 4.4 ± 0.5 cm.

The overall average distance to the MN in the trans-FCR approach was 3.5 ± 1.2 mm with a Kruskal–Wallis $P < .001$. When compared to the other standard techniques, the trans-FCR

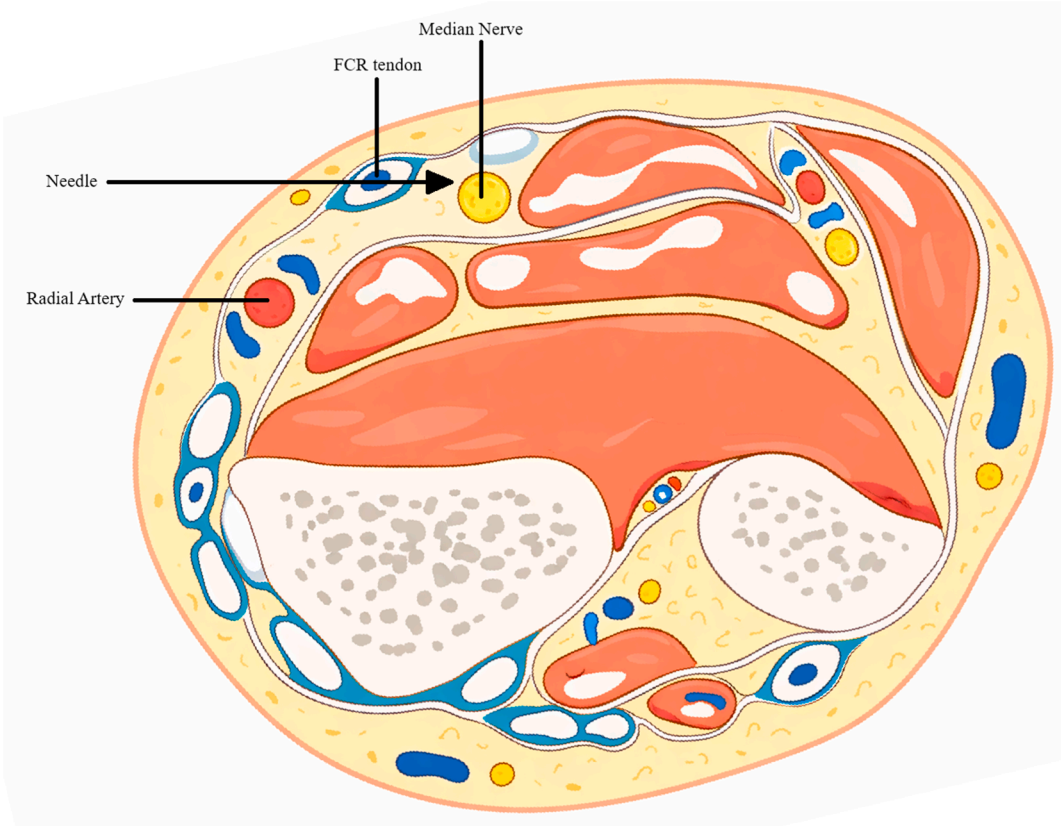


Figure 1. Illustrated cross-section of the trans-flexor carpi radialis approach. The trans-FCR needle trajectory is shown in relation to nearby critical structures. The needle traverses the FCR tendon (blue) to enter the space adjacent to the median nerve (MN). The radial artery (RA) and MN are shown as critical structures to avoid in this approach. The oval shape of the wrist allows the RA to fall dorsal and avoid penetration while the wrist is maintained in a neutral position.

approach had no instances of MN penetration (Table 2). Standard technique 1 had four instances of MN penetration. In contrast, standard technique 2 had two instances of MN penetration and two instances of PCBMN penetration. There was no significant difference ($P = .103$) in MN penetration rates among all three needle insertion techniques. The overall average distance to the MN in the trans-FCR approach was 3.5 ± 1.2 mm (range 2.0–6.8 mm). Distances to all critical structures had Kruskal–Wallis $P < .001$ across the three techniques. Two instances (14%) of PCBMN penetration occurred only in the standard 2 technique. Average distances and incidence of critical structure penetration were also stratified by presence of PL (Table 3).

Discussion

In this cadaver study, the trans-FCR technique provides a reproducible, anatomically based method for MN block under local anesthesia using the consistently identifiable FCR tendon as the primary landmark. In contrast to other traditional landmark-based approaches, the absence of the PL tendon did not impair our ability to perform the block, suggesting that the trans-FCR technique may be particularly useful in patients in whom conventional landmarks are absent or difficult to palpate.

A review of techniques between 1980 and 2026 indicated that most MN injection methods employed a shallow needle angle approximately 1 cm proximal to the distal wrist crease to deposit local anesthetic superficially around the nerve (Table 4).[17–28] Seven of the 12 techniques summarized in Table 4 rely on the PL tendon as the primary injection landmark, whereas five techniques use the FCR tendon location or a percentage of wrist diameter. Only one technique, described by Dubert and Racasan (2006),[24] specifically recommends entering through the FCR tendon. In their description, the needle is inserted through the FCR tendon at a 45° angle, 1 cm proximal to the wrist crease. However, cross-sectional wrist anatomy suggests that this trajectory may traverse the FCR tendon and subsequently the index-finger FDS tendon, the most radial of the FDS tendons, before contacting the radius, rather than entering the MN region. In contrast, our trans-FCR approach uses a perpendicular trajectory through the FCR tendon at a more proximal level, prior to branching of the PCBMN and at the musculotendinous crossover where the MN emerges more superficially.

Several anatomic features of the distal forearm make the FCR tendon an attractive tactile landmark for MN injection. The FCR tendon crosses the wrist at the scaphoid tubercle and inserts on the base of the second metacarpal; the scaphoid tubercle is typically easily palpated, even in patients with larger body habitus that may obscure other landmarks. Importantly, our technique does not depend on the PL tendon, whose presence is highly variable, with a reported prevalence ranging from 1.5% to 63.9% across different populations.[16] In diverse patient cohorts, reliance on an anatomically variable structure may reduce the reliability of landmark-based nerve blocks, whereas the FCR tendon provides a more consistent guide.

Another advantage is that the FCR tendon has its own sheath in the superficial forearm, which provides distinct tactile

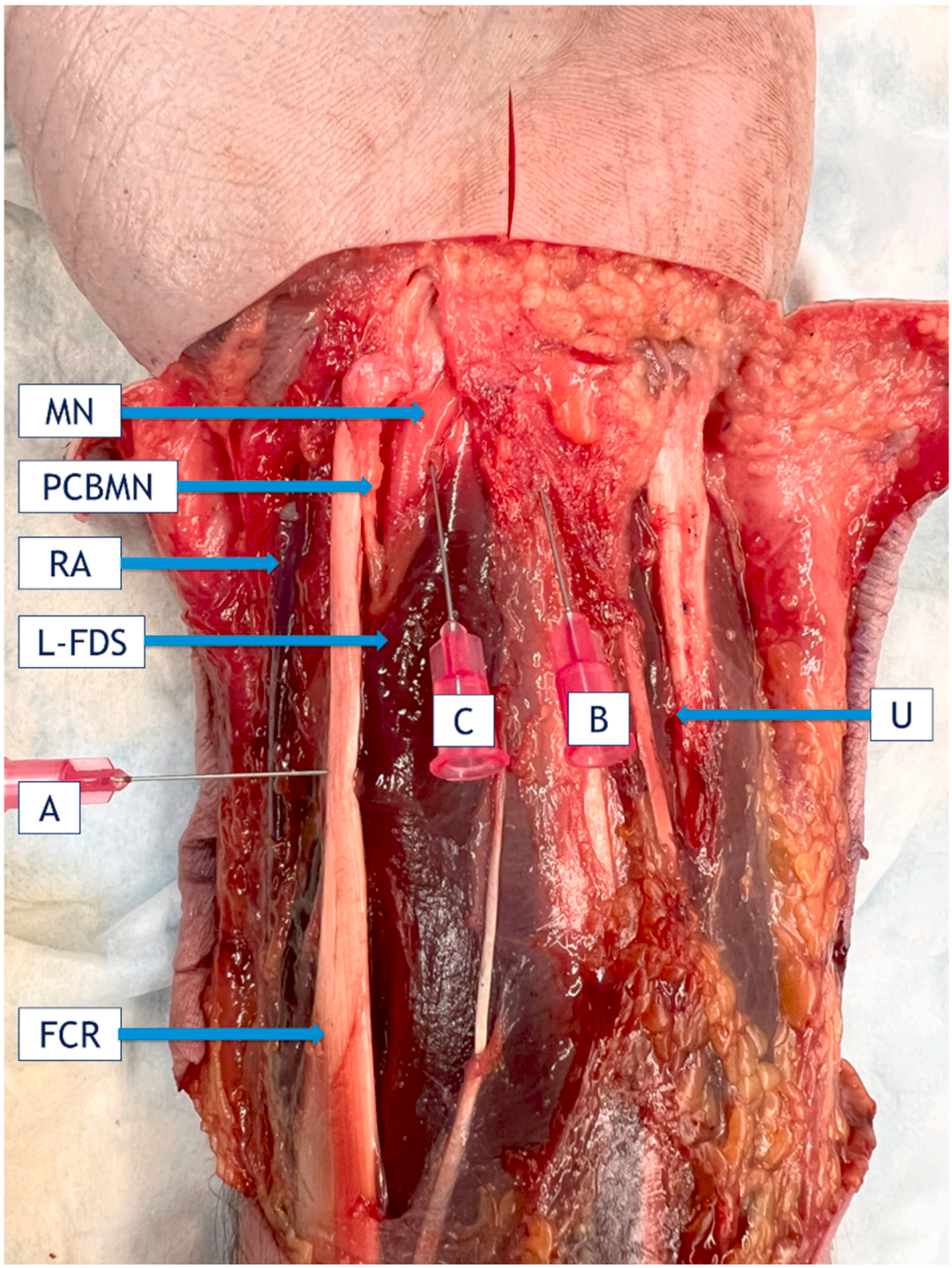


Figure 2. Cadaver dissection of the trans-flexor carpi radialis approach. The trans-FCR approach **A** compared with two other common MN local block techniques, in line with the fourth digit technique **B** and the 1 cm proximal to one-third the width of the wrist crease technique **C**. A 25-gauge needle was used to demonstrate the technique and the approximate location of anesthetic deposition. Ulnar structures (U), including the ulnar artery and nerve, are exposed to demonstrate how much farther they are out of the way from harm. FCR, flexor carpi radialis; L-FDS, long finger flexor digitorum superficialis; MN, median nerve; PCBMN, palmar cutaneous branch of the median nerve; RA, radial artery.

feedback as the needle traverses the tendon and sheath just radial to the MN. If the needle is inserted too superficially, there would be infiltration of the FCR sheath, and the operator may then subsequently advance the needle to pop through the far side of the FCR tendon sheath. There is potential to end up superficially in the antebrachial fascia if the needle is aimed volarly during insertion through the FCR. In our dissections, we identified a discrete cleft at the level where the MN passes underneath the long finger FDS to become more superficial. At approximately 4 cm proximal to the wrist crease, the long finger FDS muscle belly forms a superficial barrier, whereas the MN crosses at this depth, creating a potential fibrous space well suited for local anesthetic

Table 1
Proximity of Critical Structures in Trans-FCR Median Nerve Local Technique

Palmaris Longus	Distance to MN (mm)	SD	Distance to RA (mm)	SD	Distance to PCBMN (mm)	SD
PL-	2.7	0.6	7.6	2.4	6.4	5.5
PL+	3.8	1.2	8.7	3.8	3.0	1.4
Overall	3.5	1.2	8.5	3.5	3.8	2.9
P value	.171		.661		.072	

Note: Average proximity of structures in trans-FCR technique in millimeters. Measurements were stratified by presence of palmaris longus (PL). Average distances from the needle insertion to the median nerve (MN), radial artery (RA), and palmar cutaneous branch of the median nerve (PCBMN) are tabulated with the SD. There was no significant difference in distance to critical structures based on presence or absence of the PL tendon.
PL-, anatomic specimen without PL; PL+, anatomic specimen with PL.

Table 2
Proximity of Critical Structures in Various Median Nerve Local Anesthesia Techniques

Technique	Distance to MN (mm)	SD	Distance to RA (mm)	SD	Distance to PCBMN (mm)	SD
Trans-FCR	3.5	1.2	8.5	3.5	3.8	2.9
Standard 1	5.2*	4.2	18.5	5.5	10.7	3.2
Standard 2	3.0*	2.1	7.2	4.2	2.1*	1.5
P value	<.001		<.001		<.001	

Note: Mean proximity of structures in trans-FCR technique and standard approaches. Standard 1 represents insertion of the needle longitudinally along the fourth ray, 1 cm proximal to the wrist crease. Standard 2 represents insertion of the needle longitudinally at one-third the diameter of the wrist in 30° to the horizontal at a location 1 cm proximal to the wrist crease. Mean distances from the needle insertion to the median nerve (MN), radial artery (RA), and palmar cutaneous branch of the median nerve (PCBMN) are shown with the SD. Distances to the MN, RA, and PCBMN differed significantly among the techniques.
* Indicates penetration of the neurovascular structure using the technique. Standard technique 1 had four instances of MN penetration. Standard technique 2 had two instances of MN penetration and 2 instances of PCBMN penetration. Only nerves sustained penetration injuries.

Table 3
Proximity of Critical Structures in Various Median Nerve Local Anesthesia Techniques Grouped by Presence of Palmaris Longus

PL-	Distance to MN (mm)	SD	Distance to RA (mm)	SD	Distance to PCBMN (mm)	SD
Trans-FCR	2.7	0.6	7.6	2.4	6.4	5.5
Standard 1	6.0	2.7	23.0	7.3	14.3	4.3
Standard 2	2.8*	1.6	5.9	4.1	3.2*	3.8
P value	.183		.140		.253	
PL+						
Trans-FCR	3.8	1.2	8.7	3.8	3.0	1.4
Standard 1	5.0*	4.7	17.5	5.7	9.9	2.7
Standard 2	3.1*	2.3	7.5	4.4	1.9*	1.6
P value	.195		.001		<.001	

Note: Average proximity of structures in trans-FCR technique and standard approaches. Average distances from the needle insertion to the median nerve (MN), radial artery (RA), and palmar cutaneous branch of the median nerve (PCBMN) are shown with the SD.
* Indicates penetration of a neurovascular structure. There were four instances of MN penetration with standard technique 1 in specimens with PL+. Standard technique 2 had one instance of MN penetration in both PL+ and PL− groups. Standard technique 2 also had one instance of PCBMN penetration in both PL+ and PL− groups. There was no incidence of RA penetration.

infiltration adjacent to, but not within the nerve. Typically, with more distal approaches, the maximum of local anesthesia that can be injected is usually around 5–10 mL before experiencing considerable resistance at the syringe. In contrast, the trans-FCR clinically uses this large pocket at the cross over section for a large volume of local anesthetic (20 mL) that can comfortably be deposited into this region with no resistance with the possibility of the local settling more proximally as the hand is brought to the body. Targeting this cross over area offers two main potential advantages by depositing anesthetic at or just proximal to the branching of the palmar cutaneous nerve, the trans-FCR technique may provide anesthesia to both the volar palm and the radial digits. Because the injection is performed in the relatively unconstrained distal forearm, before the nerve enters the rigid confines of the carpal tunnel where it has less mobility, the risk of inadvertent intraneural injection with over insertion of the needle is reduced.

Despite these theoretical advantages, the transtendinous nature of the trans-FCR technique imposes important limitations. It may not be appropriate for steroid injection into the carpal tunnel because of the theoretical risk to collagen integrity and tendon rupture at the wrist. For therapeutic steroid injections, ultrasound-guided or non-transtendinous landmark-based techniques remain preferred. Ultrasound allows direct visualization of the MN, PCBMN, and adjacent vascular structures and enables confirmation of extraneural needle position before injection. Recent clinical studies have shown that ultrasound-guided steroid injections and hydrodissection for carpal tunnel syndrome can improve injection accuracy and clinical outcomes compared with landmark-based techniques alone.[29,30] Although ultrasound-guided injections are increasingly considered the standard where equipment and expertise are available. Access may be limited in some emergency departments and resource-constrained settings. In such contexts, a purely tactile method such as the trans-FCR technique may provide a cost effective and readily deployable alternative to anesthetize the radial hand in the emergency department or small procedure room.

This anatomic cadaver study has inherent limitations because it may not represent the same dynamic soft tissue conditions of a live patient. All measurements were conducted with the forearm supinated and the wrist in a neutral position. Ulnar deviation of the wrist causes the MN to translate radially under the FCR.[31] The needle had to be inserted to a depth that allowed its position to be maintained during dissection, potentially resulting in deeper insertion and contributing to the higher perforation rates observed in the control group. Clinically, traversing the sensory fiber dense FCR sheath may cause pain, whereas passage through a healthy FCR tendon is typically less painful because it is sparsely innervated.[32] In contrast, stimulation of the dense innervated MN can result in sharp pain down the fingers. Additional clinical studies are needed to validate the efficacy and safety of the trans-FCR technique in a broad patient population.

Taken together, these cadaveric findings suggest that the trans-FCR technique is a reasonable anatomical approach for MN block under local anesthesia and is comparable to other landmark-based techniques. By using the consistent position of the FCR, it may reduce variability in landmark identification, particularly in patients without a PL, while posing minimal risk of iatrogenic injury to critical neurovascular structures. Needle depth can be readjusted to the appropriate depth based on infiltration of the FCR tendon sheath. Although this tactile based technique makes it attractive for resource limited environments as a potential adjunct, the trans-FCR technique should currently be regarded solely as an anatomically validated option. Prospective clinical studies, with comparisons to both traditional landmark-based and ultrasound-guided techniques, will be needed to define its role in routine care.

Table 4
Historically Described Techniques to Avoid Iatrogenic Injury for Median Nerve Injection, for Local Anesthetic (1980–2026)

Author(s)	Year	Anatomic Location
Gelberman et al[17]	1980	1 cm proximal to the distal wrist crease between the PL and FCR with 45° to 60° angle
Green[18]	1984	Ulnar to PL
Linskey and Segal[19]	1990	Ulnar to PL tendon at the distal wrist crease If not present, injection in line with the fourth digit
Frederick et al[20]	1992	Midway between PL and FCU, just proximal to the wrist crease If PL is not present, inject in line with the FDS tendon of the fourth digit
Kasten and Louis[21]	1996	Distal and dorsal direction, start on the radial border of the pisiform toward the mid portion of the carpal tunnel
Dammers et al[22]	1999	Volar side of the forearm between PL tendon and FCR tendon, 4 cm proximal to the wrist crease
Graham et al[23]	2004	Ulnar to FCR
Dubert and Racasan[24]	2006	Through FCR tendon, 1 cm proximal to the wrist crease with the needle angled at 45° toward the medial edge
MacLennan et al[25]	2009	Ulnar to FCR tendon at the wrist crease
Kamanli et al[26]	2011	Ulnar to PL tendon distal to the wrist crease
Menge et al[27]	2016	30% to 33% of wrist width from the radial styloid
Brooks et al[28]	2019	Radial to PL and midline or ulnar to FCR, between 8 and 10 mm radial to the midpoint of the wrist

Note: This table summarizes various techniques described over the past four decades to minimize iatrogenic injury during median nerve (MN) injections. The authors and years of publication are listed alongside the specific anatomic locations recommended for injection. Techniques have evolved from approaches based on the palmaris longus (PL) and flexor carpi radialis (FCR) tendons to approaches that account for anatomic variation, including absence of the PL tendon.

Conflicts of Interest

No benefits in any form have been received or will be received related directly to this article.

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